

Nicola Zomer

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🔗 NicolaZomer

in nicolazomer

Experience

Marine Weather Intelligence (MWI), Remote (FR)

French deeptech startup building AI-driven solutions to improve routing, weather forecasting, and maritime safety.

PhD Student

Aug 2026 – Present

Advisor: Dr. Valentin Lefranc

- Progressively integrating research outcomes into MWI's operational weather routing pipeline.

Research Engineer

Sep 2024 – Aug 2026

- Developed, optimized, and deployed core weather routing algorithms powering MWI's commercial platforms in Python, Rust, and C++, achieving 10× speedup and 50% memory reduction.
- Developed a proof-of-concept for maritime route optimization under meteorological uncertainty that secured the BPI France i-Lab innovation grant (project WUMRO).
- Tools: FastAPI, Docker, Git, Grafana, Dockploy.

University of Padua, Padua (IT)

Aug 2023 – Apr 2024

Master Thesis: "Swarm Intelligence in Interacting Language Model Agents"

Supervisor: Prof. Manlio de Domenico (Complex Multilayer Networks Lab)

- Extended DeepMind's OPRO to a multi-agent framework, improving convergence on complex optimization tasks.
- Investigated emergent collective behaviors in networks of LLM agents via the Schelling model of segregation.
- Published in *npj Artificial Intelligence* (2026).

Central European University, Vienna (AT)

Apr 2023 – Jun 2023

Research Intern

Supervisor: Prof. Tiago P. Peixoto (Inverse Complexity Lab)

- Applied generative network models and Bayesian inference to model platform behavior in the Fediverse, analyzing 4.2M scraped data points.
- Developed an ad-hoc MM algorithm for maximum likelihood estimation with ~50k parameters.

Education

PhD in Computer Science

Aug 2026 – Present

Sorbonne University (FR)

Supervisors: Prof. Giulia de Masi and Prof. Antonio Celani

- Thesis: *Data-driven ship route optimization under dynamic and uncertain environments.*
- Co-designed the doctoral project with MWI, identifying potential supervisors and formulating the initial research plan.

M.Sc. in Physics of Data

Oct 2021 – Apr 2024

University of Padua (IT) – Final Grade: 110/110 with honors (GPA: 29.9/30)

- Relevant courses: Laboratory of Computational Physics, Management and Analysis of Physics Datasets, Machine Learning, Neural Networks and Deep Learning, Information Theory and Inference.

B.Sc. in Aerospace Engineering

Sep 2016 – Jul 2019

University of Padua (IT) – Final Grade: 110/110 with honors (GPA: 29.1/30)

- Extracurricular: Mission Analysis Engineer at *Alba CubeSat UniPD*, selected for ESA's Fly Your Satellite program.

Honors

"Mille e una lode" Scholarship (A.Y. 2018/2019, 2019/2020, 2020/2021). Awarded by the University of Padua to the top 3% students of each degree program, based on GPA.

Lead the Future Mentorship (LTF). Selected to be a mentee for LTF, a leading mentorship non-profit organization for students in STEM, with an acceptance rate below 20%.

Bronze Medal, Italian Mathematical Olympiad in Teams, 2018. National final, top 32 out of over 600 teams.

Publications

Zomer, N. & De Domenico, M. **Unraveling the emergence of collective behavior in networks of cognitive agents.** *npj Artificial Intelligence*, 2026. [DOI]

Bertè, M., Ibrahimli, R., Koopmans, L., Valgañón, P., Zomer, N. & Colombi, D. **Enhancing stop location detection for incomplete urban mobility datasets.**, 2024. [arXiv]

Academic Service

Peer Reviewer, IEEE Journal of Oceanic Engineering (2026 – Present).
Local Organizing Committee Member, MARIS-AI Symposium, Abu Dhabi (2027).

Languages

Italian: Native | English: Fluent (C1) | Spanish: Intermediate (B1) | French: Basic (A2)